

LESSON 6.1

PROPORTIONAL RELATIONSHIPS



LESSON INTRODUCTION

MA.8.AR.3.1 Determine if a linear relationship is also a proportional relationship.

LEARNING TARGETS

- I can describe attributes of a proportional relationship between two variables represented by tables and equations.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.4.1 Determine whether two quantities have a proportional relationship by examining a table, graph, or written description.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How can you determine if a relationship is a proportional relationship?
- How can you represent a proportional relationship using an equation?

LESSON NARRATIVE

In this lesson, students explore proportional relationships in real-world situations. Students initially review what makes a set of ratios proportional. Then, students analyze what makes a proportional relationship in graphs, tables, and equations.

COMPONENT SUMMARY

6.1.1 WARM-UP (~5 MIN.)

Reflect on importance of problem-solving skills for future jobs

6.1.2 EXPLORATION (10-15 MIN.)

Explore proportional relationships

6.1.3 GUIDED INSTRUCTION (20-25 MIN.)

Determine what makes a relationship proportional

6.1.4 WRAP-UP (~5 MIN.)

Explain how to recognize a proportional relationship in an equation and table

RESOURCES

GLOSSARY

Key Terms:

- Proportional Relationship
- Independent variable
- Dependent variable
- Constant of proportionality

Additional Terms:

- Equivalent
- Ratio

MATERIALS

- N/A

ADDITIONAL PREPARATION

- Warm-Up 6.1.1 contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may have trouble distinguishing between dependent and independent variables.
- When writing an equation, students may reverse the x- and y-values.
- Students may be confused when variables other than x and y are used.

THINGS TO CONSIDER

N/A

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 6.1.2 Exploration, have students initially complete the problems independently. Then, ask students to discuss their work with a partner. Finally, debrief with the whole class. These discussions will help students gain confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students apply their knowledge of proportional relationships to real-world contexts 6.1.3 Guided Instruction. Encourage students to ask themselves to interpret what their answers mean in the given context and if their answer makes sense in the context. For example, students are asked what an x value of 0 would mean for the ladybug or ant.

DIFFERENTIATE INSTRUCTION

Student comprehension of the vocabulary is essential. Consider creating a Word Wall with definitions, example, and visuals.

6.1.1 WARM-UP: DAY AT WORK - BRIDGE BUILDER

TEACHER MOVES

Play the Career Profile Video on the work bridge builder for the entire class. After students watch the video (2:32 minutes), prompt them to reflect on the profession by answering the question independently. Consider asking students if they have considered a career. Focus on the idea that education is learning how to think and solve problems and how useful that can be in a variety of professions.



6.1.1 Warm-Up: Day at Work - Bridge Builder

Ken Brown said, "You are hammered all through school to think, and the thinking part is a big part of your education where you are taught how to think and solve problems."



1. Why is learning how to think and solve problems important for any job you do in the future?

Answers will vary. Employers like to see good problem-solving skills to show that you have a range of competencies that are useful in the workplace. Problem solvers can come up with solutions to identified problems and can make decisions with confidence.



6.1.2 Exploration: What is a Proportional Relationship?

The table below shows how many cans of disinfectant spray a hospital receives depending on the number of cases ordered.

Cases Ordered	Number of Cans
1	12
3	36
5	60
10	120

- How many cans are in each separate case when 3 cases are ordered?
There are 12 cans in each separate case.
- How many cans are in each case when 5 cases are ordered?
There are 12 cans in each case.
- How many cans are in each case when 10 cases are ordered?
There are 12 cans in each separate case.
- Explain how to determine the number of cans of disinfectant spray that would be in each separate case when ordering any number of cases.
Answers will vary. There are 12 cans for every 1 case. The unit rate is 12:1. The number of cans divided by the number of cases ordered always equals 12.
- This situation represents a proportional relationship. Discuss with your partner why the situation is a proportional relationship and summarize your conclusions.
Answers will vary. Each time the cases ordered is multiplied by 12 to get the total number of cans. It is a proportional relationship because the same value is multiplied each time. Each ratio of the number of cans to the number of cases is 12:1. It is proportional relationship because each ratio is equivalent.

6.1.2 EXPLORATION: WHAT IS A PROPORTIONAL RELATIONSHIP?

TEACHER MOVES

This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

ENRICHMENT

Notice and Wonder

What do you notice about the number of cans of disinfectant per case? Why do you think this is? What do you wonder?

TEACHER MOVES

Students may not use vocabulary like ratio or unit rate. Having students compare their reasoning for why the situation is proportional with other students in the class can increase precision with the use of these vocabulary terms for students.

6.1.3 GUIDED INSTRUCTION: UNDERSTANDING PROPORTIONAL RELATIONSHIPS

TEACHER MOVES

Students learn about ratios in grade 6 and extend that learning to proportions in grade 7. This lesson is designed to spiral through the concepts that students should have solidified in these grades. Allow students to have some productive struggle by pausing during your instruction. Give students time to recall the content. The lesson is scaffolded to help rebuild student understanding. Use the additional questions as needed.

ENRICHMENT

- What does equivalent ratios mean?
- How can you check if equivalent ratios exist?

DIFFERENTIATE INSTRUCTION

Student comprehension of the vocabulary is essential. Consider creating a Word Wall with definitions, examples, and visuals.

ENRICHMENT

How do you determine which quantity is the dependent variable and which is the independent variable?

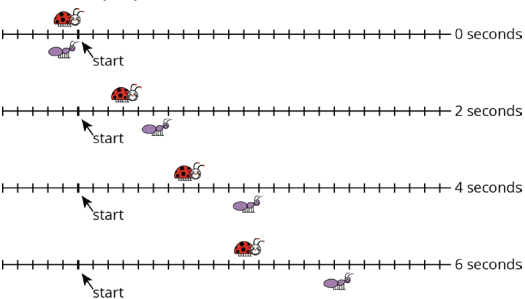


6.1.3 Guided Instruction: Understanding Proportional Relationships



A **proportional relationship** is a collection of pairs of numbers that are in equivalent ratios.

A ladybug and an ant move at constant speeds. The diagram with tick marks shows their positions at different times, in seconds (sec). Each tick mark represents 1 centimeter (cm).



1. Complete the table of values for the ladybug and the ant.

Ladybug	
Time (sec)	Distance (cm)
0	0
2	4
4	8
6	12

Ant	
Time (sec)	Distance (cm)
0	0
2	6
4	12
6	18

One way to determine if a relationship is proportional is to identify if **equivalent** ratios exist.

When writing ratios for situations with two variables, each ratio should compare the dependent variable to the independent variable and can be written in fractional form as $\frac{\text{dependent variable}}{\text{independent variable}}$.

The **independent variable** is the quantity in the relationship whose values do not depend on changes in other variables. It is typically represented by the variable x .

The **dependent variable** is the quantity in the relationship whose values depend on changes in the independent variable. It is typically represented by the variable y .

2. Time and distance are the two variables in this situation. Complete the ratio identifying the dependent and independent variables.
$$\frac{\text{dependent}}{\text{independent}} = \frac{\text{distance}}{\text{time}}$$
3. Complete the set of ratios for each insect and determine if the ratios within each set are equivalent.

Ladybug	
$\frac{4}{2} = 2$	
$\frac{8}{4} = 2$	
$\frac{12}{6} = 2$	
Are the ratios equivalent?	yes

Ant	
$\frac{6}{2} = 3$	
$\frac{12}{4} = 3$	
$\frac{18}{6} = 3$	
Are the ratios equivalent?	yes

4. Explain how you know if the tables each represent a proportional relationship.
Answers will vary. I know they are proportional because all of the ratios are equivalent. I know they are proportional because they have the same constant of proportionality.
5. Discuss with your partner why the ratio, when time is equal to zero, is not determined for each situation and summarize your conclusions.
Answers will vary. When the time is zero, the ant and ladybug have not moved, so the distance is also zero. The time starts when the bugs begin to move.



The **constant of proportionality**, k , is the constant value of the ratio of two proportional quantities.

6. The equation $y = kx$ can be used to represent any proportional relationship. Use the word bank to label each part of the equation.

independent variable	dependent variable	constant of proportionality
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constant of proportionality

dependent variable $y = k \cdot x$ *independent variable*

- a. What is the constant of proportionality for the ladybug?
2
- b. Complete the table to explain what the constant of proportionality, k , and the x - and y -variables represent in the context of the ladybug's movement.

Value	Meaning in Context
k	<i>2 cm per second</i>
x	<i>time passed</i>
y	<i>total distance</i>

6.1.3 GUIDED INSTRUCTION:
UNDERSTANDING PROPORTIONAL
RELATIONSHIPS (CONTINUED)

TEACHER MOVES

Think-Pair-Share

Have students initially complete the problems independently. Then, ask students to discuss their work with a partner. Finally, debrief with the whole class. These discussions will help students gain confidence and understanding.

DIFFERENTIATE INSTRUCTION

Consider reviewing the formula $k = \frac{y}{x}$ for the constant of proportionality. Consider also posting it on the board or somewhere students can view it.

6.1.3 GUIDED INSTRUCTION: UNDERSTANDING PROPORTIONAL RELATIONSHIPS (CONTINUED)

ENRICHMENT

- Which insect had the larger constant of proportionality?
- What does that mean in the context of the situation?

ENRICHMENT

- What does the equation tell us about the ladybug's movement?
- What does the equation tell us about the ant's movement?

DIFFERENTIATE INSTRUCTION

This idea may be challenging for some students. Consider talking about the properties of 0 and what it means in the context of the situation.

- Write the equation for the ladybug's movement.
 $y = 2x$
- What is the constant of proportionality for the ant?
3
- Complete the table to explain what the constant of proportionality, k , and the x - and y -variables represent in the context of the ant's movement.

Value	Meaning in Context
k	3 cm per second
x	time passed
y	total distance

- Write the equation for the ant's movement.
 $y = 3x$
 - Discuss with your partner if it is possible to write an equation for a proportional relationship if only the constant of proportionality is given. Summarize your discussion below.
Answers will vary. It is possible because we know the equation for a proportional relationship is $y = kx$ with k being the constant of proportionality.
 - Use the equation for the ladybug's movement to determine the value of y when $x = 0$.
 $y = 2x$
 $y = 2 \cdot 0$
 $y = 0$
 - Use the equation for the ant's movement to determine the value of y when $x = 0$.
 $y = 3x$
 $y = 3 \cdot 0$
 $y = 0$
7. Use the equation $y = k \cdot x$ to explain why the value of y will be 0 when $x = 0$ no matter the value of the constant of proportionality, k .
Answers will vary. When multiplying any number by zero, the product will be zero.



6.1.4 Wrap-Up: Cool Down

1. A textbook stated, “We say that y is directly proportional to x if $y = kx$ for some constant k .”

For homework, students were asked to restate the definition in their own words and give an example. Below is the response from one student.

Jessica	When two quantities are proportional, it means that as one quantity increases the other will also increase at the same ratio for all values. An example is the circumference of a circle and its diameter. The ratio of the values always equals π .
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Write your own definition and example, including how to recognize a proportional relationship from a table and an equation.

Name	Definition & Example
	<i>Answers will vary. If we multiply the x value by the same number or the constant to achieve the y value, then the relationship is proportional. The equation $y = kx$ demonstrates this relationship.</i>

6.1.4 WRAP-UP: COOL DOWN

DIFFERENTIATE INSTRUCTION

Some students may struggle with expressing themselves in writing. Consider providing alternatives, such as using speech-to-text software, an audio recorder (i.e. Flipgrid), or labeling examples.